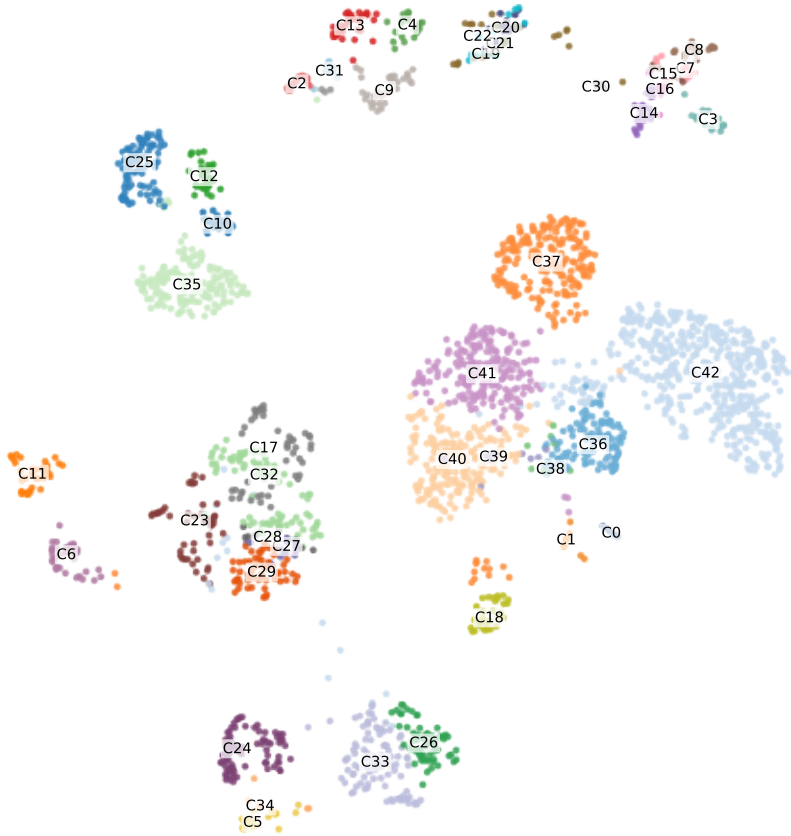


Adult pancreas scRNA benchmark
TBS adaptive diffusion recomputed-NNLS branch-time (k90, alpha=0.01, edge=0.001)

selected NNLS-TBS clusters



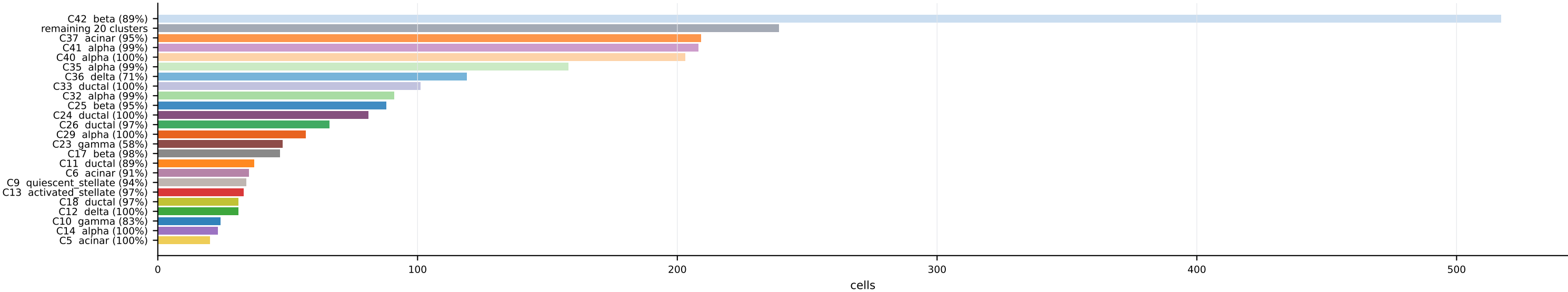
real full tree



cells / clusters: 2,500 / 43
ARI: 0.413
V-measure: 0.662
cluster purity: 0.929
dominant-label recall: 0.484
NNLS R2 / corr.: 0.974 / 0.987
NNLS RMSE / MAE: 0.307 / 0.201

all 2,500 leaves; radius = cumulative NNLS branch time; colors match the labeled UMAP

largest clusters and dominant reference label

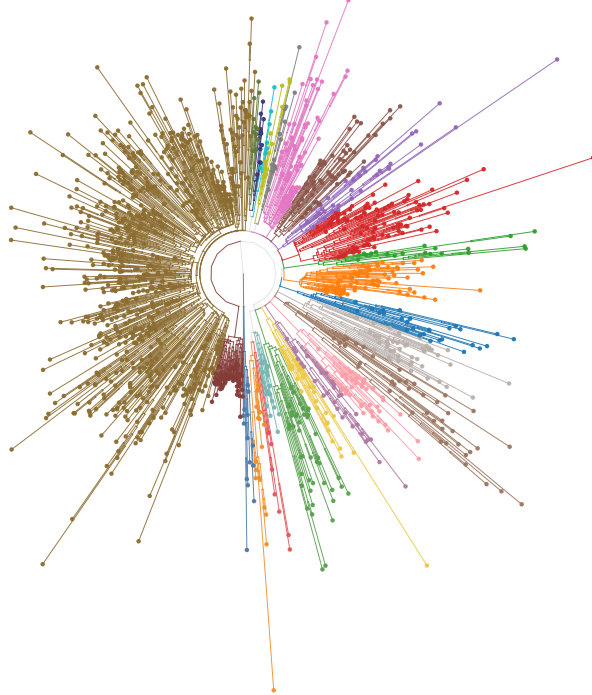


Goncalves fetal pancreas progenitor benchmark
TBS adaptive diffusion recomputed-NNLS branch-time (k90, alpha=0.01, edge=0.001)

selected NNLS-TBS clusters



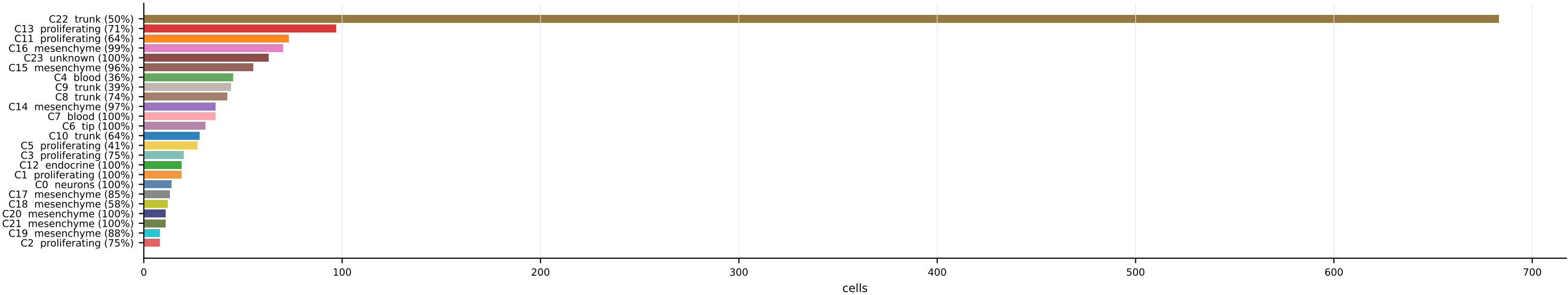
real full tree



cells / clusters: 1,465 / 24
ARI: 0.206
V-measure: 0.410
cluster purity: 0.655
dominant-label recall: 0.540
NNLS R2 / corr.: 0.807 / 0.899
NNLS RMSE / MAE: 0.335 / 0.254

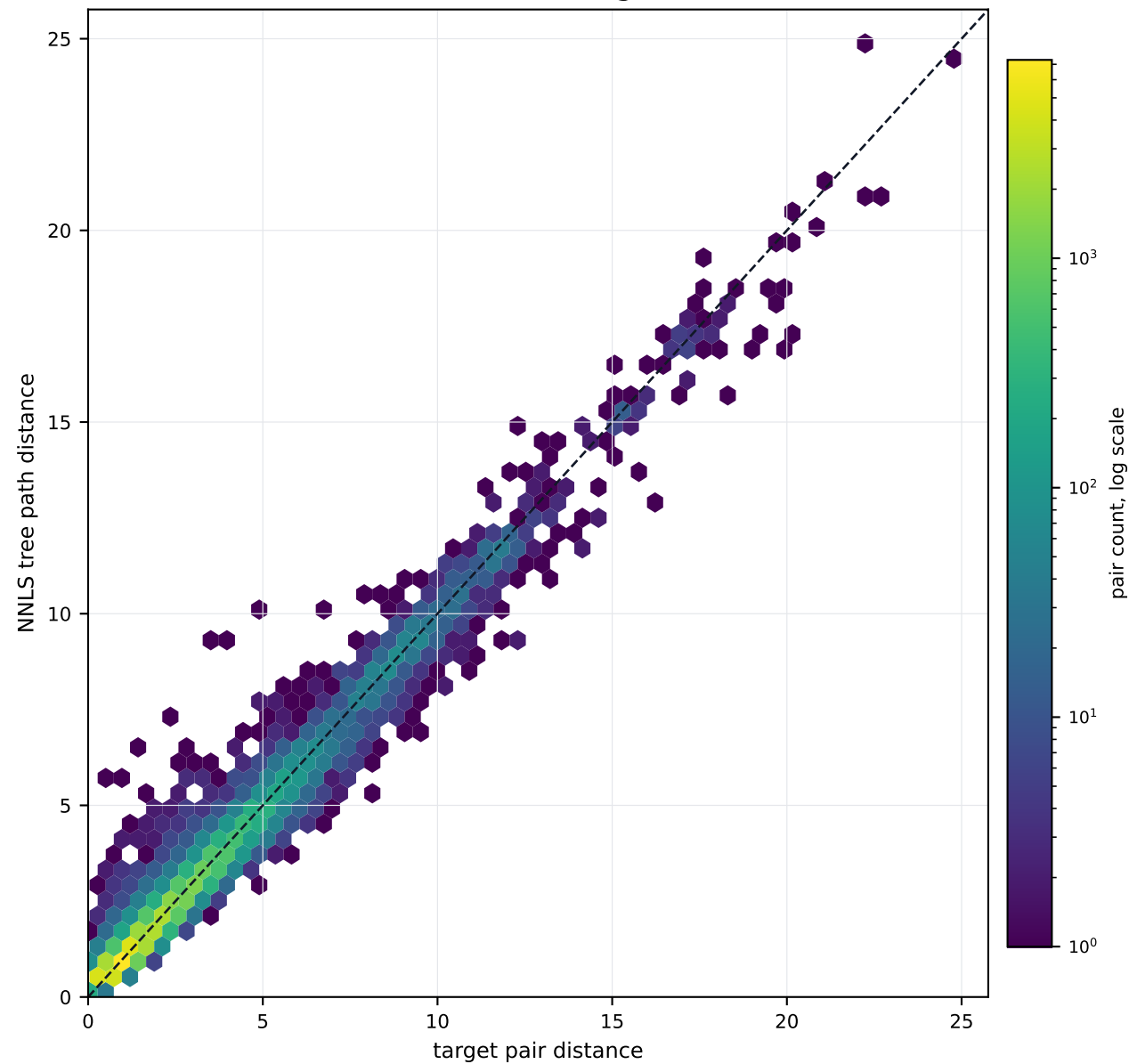
all 1,465 leaves; radius = cumulative NNLS branch time; colors match the labeled UMAP

largest clusters and dominant reference label



Same NNLS fit diagnostic for both analyses

Adult pancreas scRNA benchmark
NNLS branch-length fit



Goncalves fetal pancreas progenitor benchmark
NNLS branch-length fit

